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Department of Transportation

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14 CFR Part 23

**Airworthiness Standards; Fatigue
Requirements; Final Rule**

DEPARTMENT OF TRANSPORTATION

Federal Aviation Administration

14 CFR Part 23

[Docket No. 25086, Amendment No. 23-38]

RIN 2120-AA58

Airworthiness Standards; Fatigue Requirements

AGENCY: Federal Aviation Administration (FAA), DOT.

ACTION: Final rule.

SUMMARY: This amendment adopts fatigue requirements for empennage, canard, tandem wing, and winglet/tip fin structures for airplanes certificated in the normal, utility, acrobatic, or commuter category. Service history, airworthiness directives and field reports related to unsafe, or potentially unsafe, fatigue failures of empennage structural components on airplanes of these types have shown the need to consider empennage fatigue strength during type certification. Due to recent developments in airplane designs in which conventional wing and empennage configurations are being replaced or modified by canard, tandem wing and winglet/tip fin structures which also are essential for safe flight, fatigue requirements for these new structures are included. This amendment establishes a level of safety for the prevention of catastrophic fatigue failures of these structures.

EFFECTIVE DATE: October 26, 1989.

FOR FURTHER INFORMATION CONTACT: Earsa L. Tankesley, Standards Office (ACE-110), Aircraft Certification Division, Central Region, Federal Aviation Administration, Room 1656, 601 East 12th Street, Kansas City, Missouri 64106; telephone (816) 426-5688.

SUPPLEMENTARY INFORMATION:

Background

Service history, airworthiness directives, and documented field reports (e.g., malfunction or defect reports and maintenance difficulty records) related to unsafe, or potentially unsafe, fatigue/vibration conditions of empennage components on small airplanes show the need to consider empennage fatigue as part of the type certification process. Unique design features, such as canards, tandem wings, and winglets/tip fins are being incorporated into the designs of airplanes certificated to the requirements of part 23. These airplanes typically operate in relatively severe fatigue environments with short time/distance flights made at comparatively

low altitudes. The increased utilization of these airplanes in commuter and air taxi operations has resulted in these airplanes being subjected to increased exposure to loads that cause fatigue damage to the structure, thereby increasing the potential for fatigue related failures.

The FAA is continuing to consider comments on definitions of certain terms that are relevant to this rule. See Small Airplane Airworthiness Review Program Notice 2, Notice of Proposed Rulemaking (NPRM) (54 FR 9276; March 6, 1989). When final action, if any, is taken as a result of that NPRM and the definitions are incorporated into Part I, those definitions will also apply to part 23. For ease of reference, the proposed definitions are included below.

"Canard" means the forward wing of a canard configuration which may be a fixed, movable, or variable geometric surface, with or without control surfaces.

"Canard configuration" means a configuration in which the span of the forward wing is substantially less than that of the aft wing.

"Tandem wing configuration" means a configuration having two wings of similar span, mounted in tandem.

"Winglet or tip fin" means an out-of-plane surface extending from a lifting surface. This surface may or may not have control surfaces.

"Forward wing" means a forward lifting surface of a canard configuration or tandem-wing configuration airplane. The surface may be a fixed, movable, or variable geometric surface, with or without control surfaces.

Empennage fatigue requirements have been in effect since 1979 for those propeller-driven, multiengine small airplanes that were type certificated in accordance with Special Federal Aviation Regulations (SFAR) 41. Although wing fatigue regulations for small airplanes certificated to part 23 have been in effect since amendment 23-7 (34 FR 13078; August 13, 1969) similar regulations for critical empennage structures of small airplanes do not exist in part 23.

During recent years, the size, weight, power, and airspeed ranges of small airplanes have increased significantly. Generally, these factors have a negative effect on the fatigue characteristics of the empennage structure. Failures of empennage components have resulted in fatal airplane accidents, near-fatal accidents, severe operational problems for pilots, and disturbing experiences for passengers. Similar failures of canard or forward-wing structures are likely to occur if these structures are not evaluated for fatigue strength.

As a result of in-service empennage fatigue/vibration structural failures which have occurred in the past few years, over three dozen airworthiness directives have been issued. Additionally, several hundred service reports, which relate to empennage fatigue problems, have been submitted by field personnel (pilots, mechanics, and inspectors) as part of the FAA's Malfunction or Defect (M & D) Program.

On February 15, 1987, new commuter category airplane type certification requirements were added to part 23 with amendment 23-34 (52 FR 1806; January 15, 1987). Although this amendment added requirements to § 23.572 which require fatigue evaluation of the vertical fin, horizontal stabilizer, and attachment structures for commuter category airplanes, no similar requirements were added for airplanes certificated in the normal, utility, or acrobatic categories. Furthermore, the fatigue requirements adopted did not specifically address canards, tandem wings, or winglets/tip fins.

As a result of the above-discussed empennage fatigue history and airplane design trends, the FAA issued notice 86-14 (51 FR 33700; September 22, 1986) to amend § 23.572. At the time notice 86-14 was issued, amendment 23-34 has not been incorporated into the FAR; therefore, no mention is made of amendment 23-34 in the notice 86-14. However, the FAA anticipated the incorporation of amendment 23-34, and notice 86-14 was developed to include all airplanes certificated to the airworthiness requirements of part 23. Although amendment 23-34 established separate fatigue evaluation requirements for commuter category airplanes in § 23.572(b), the regulatory action of this final rule adds substantively equivalent requirements for all other part 23 airplanes. To eliminate redundant statements of the fatigue evaluation requirements applicable to all categories of part 23 airplanes, these requirements are stated in § 23.572(a). For clarity, an editorial change is made to § 23.572(a)(1).

Since separate fatigue evaluation requirements for commuter category airplanes are no longer necessary, § 23.572(b), which was added by amendment 23-34, is removed. A new § 23.572(b) is added which specifies minimum requirements for loads spectra that are part of the fatigue evaluations required in § 23.572(a).

Discussion of Comments

Three commenters submitted responses to notice 86-14. Two commenters express support for both

the intent of the rule and the content of the proposed amendment. One of these commenters noted that optimizing the detailed design by the use of computerized stress analysis techniques and the "stretching" of basic models to accommodate increased weight and speeds have resulted in more efficient structures and a reduction in the strength margins which formerly provided inherent fatigue capabilities erosion of inherent fatigue capabilities due to the reduction in static. This commenter also noted that the increasing use of higher strength aluminum alloys has enabled working stresses to be increased, and these higher working stresses will make modern structures more prone to fatigue. For these reasons, this commenter concludes that fatigue analysis is essential for any structure, including the empennage, the failure of which could prove catastrophic.

One commenter suggests that the definition of "canard", "canard configuration", "tandem wing configuration", and "winglet", as provided in the discussion paragraph of notice 86-14, be incorporated in the FAR, preferably in part 1. The FAA has proposed such regulatory action as a result of comments received at the part 23 Review Conference. The definitions suggested by this commenter were proposed in the Small Airplane Airworthiness Review Program Notice 2 (54 9276; March 6, 1989).

One commenter states concern about the FAA meaning of "the effects of lifting surface wakes" as proposed for § 23.572(b). This commenter proposes that the FAA explain what is being alluded to and "how one goes about such an evaluation." The intent of the FAA in using the phrase, "the effects of lifting surface wakes" was to include any loading on the structure being evaluated which is the result of the aerodynamic effects from another lifting surface on the airplane and which has a significant effect on the fatigue loading of the structure being evaluated. As examples, lifting surface wake effects may include: (1) Mutual aerodynamic effects (downwash effects) which can alter the steady-state spanwise load distribution on other lifting surfaces, and (2) vortex effects from a forward-lifting surface which induce vibratory loads on the trailing surfaces. The final rule replaces the term "lifting surface wakes" with the more definitive terms "mutual influence of aerodynamic surfaces" and "buffet from vortex impingements." The subject of lifting surface wake effects was discussed at the part 23 Review Conference in October 1984, and at that

time, the FAA received recommendations to incorporate type certification requirements to account for structural loading from these effects. Although regulatory actions based on these recommendations are still under consideration, the FAA considers the influence of these loading effects to be an essential part of the fatigue evaluation of certain airframe designs.

This commenter also suggests that the proposed § 23.572(b) in notice 86-14 be revised as two sections to read as follows:

(b) Each evaluation required by this section must include typical loading spectra (e.g., taxi, ground-air-ground cycles, maneuver, gust) effects.

(c) Account must be taken of any significant contribution of propeller wake-induced vibrations when complying with this section.

This commenter contends that this is necessary because the threefold scope of the proposed § 23.572(b) does not lend itself to inclusion of finite load spectra. The FAA agrees with the concerns expressed by the commenter. The finite loading spectra used in the evaluation required by § 23.572 must include typical loading spectra for the type airplane which significantly contribute to the fatigue damage. In certain design cases, the unique sources of loading, such as propeller slipstream loading, the mutual influence of aerodynamic surfaces, or the buffet from vortex impingement, may contribute to the fatigue damage of various structural components depending on the airplane design. Any of these sources of loading, which is determined to have a significant effect on the fatigue damage of a structural component, should be considered in the fatigue evaluation. However, in many cases, fatigue damage resulting from the effects of the propeller slipstream, the mutual influence of the aerodynamic surfaces, or buffet due to vortex impingement can be avoided by acceptable design detail. Therefore, paragraph (b) is restated to ensure a clearer understanding of the rule. The changes to § 23.572(b) from the wording in notice 86-14 are made to clarify the intent of the rule and result in no substantive changes to the requirements, as proposed in notice 86-14.

One commenter opposes including fatigue requirements for the vertical surfaces unless load spectra information, or sources of this information, are included in the preamble to this amendment. This commenter questions "whether or not the published spectra data provides sufficient basis to specify a loading

spectra for the vertical surfaces." The FAA has established a project to develop more specific data on certain load conditions which are a part of the fatigue spectra for vertical surfaces. However, since 1979, type certification of multiengine airplanes to SFAR-41 has required a fatigue evaluation of the empennage, including the horizontal stabilizer, vertical fin, and attaching structure. Applicants for SFAR-41 type certificates were able to develop fatigue loads spectra which were acceptable to the FAA. The service history on SFAR-41 airplanes does not indicate any general adverse effects associated with technology used to show compliance with the empennage fatigue requirements in SFAR-41. The FAA considers this technology applicable for developing fatigue loads to meet the requirements of this amendment to part 23.

The increasing use of unconventional empennage designs that utilize T-tail or cruciform configurations, for which the fin structure supports the horizontal stabilizer, adds to the importance of adequate fin strength. For these configurations, a failure of the fin could result in a loss of all flight stability and control provided by the empennage. Adequate fatigue strength of these structural elements must be ensured to prevent catastrophic failures. The FAA considers the fatigue evaluation of these vertical surfaces to be essential for ensuring adequate strength of these structures.

On many type certification programs (e.g., derivative models, supplemental type certificates), the structural design and anticipated fatigue environment are not significantly different from those of previously certificated airplanes. The provisions of § 23.572(a) relieve the requirement for a detailed fatigue evaluation if the structure, operating stress levels, materials, and expected uses are comparable, from a fatigue standpoint, to a similar design that has had extensive satisfactory service experience. The FAA anticipates that applicants will incorporate the concept of similar design and loading environment into their fatigue evaluation whenever warranted.

Economic Evaluation

The cost of compliance with the rule will result primarily from designing and certifying new part 23 airplane empennages, including any testing that may be required. Some manufacturing parts costs as well as operating costs may also be involved.

Interviews with manufacturers provided wide ranging costs estimated

to result from both the safe life fatigue and fail safe requirements in the proposal. For example, some manufacturers claim that, in order to perform the fatigue tests, new equipment will have to be acquired. Others maintain that equipment presently used to conduct tests on the wing carry-through and/or the empennages of larger twin-engine propeller-driven airplanes could be modified to perform such tests on the empennages of small airplanes. In addition, manufacturers claim that costs will vary depending upon whether load specification standards are provided or have to be developed.

To be conservative, the FAA has averaged the manufacturers' estimated costs including and excluding new equipment. Since the rule is not intended to apply retroactively, no recertification costs have been developed. In addition to the costs cited above, other costs may result from these rules. This will be particularly true if engineering studies determine that, in order to meet the provisions of these rules, substantial changes are required in the structural elements of newly designed empennages. Some of these changes may result in reduced costs over time. For example, more resilient new generation materials may weigh less and, thus, reduce fuel costs. Also, these rules will reduce maintenance costs because fewer fatigue cracks will have to be repaired. It is impossible to estimate these effects at this time. The FAA, therefore, assumes in this analysis that these effects will offset each other. Table 1 sets forth the estimated costs that will result from the new rule.

TABLE 1.—ESTIMATED COSTS OF EMPENNAGE FATIGUE RULE

	Average costs	
	With load specs provided	W/O load specs provided
Single Engine:		
Total cost.....	\$91,788	\$102,662
Per aircraft Cost ¹	45.89	51.33
Twin Engine:		
Total cost.....	986,901	1,000,882
Per aircraft ¹	190.24	1,000.87

¹ Assuming 2,000 single-engine and 1,000 twin-engine airplanes per production run. Currently, general aviation production levels are very low, about 700 per year for single-engine models and 140 per year for multiengine models. However, the revised rule will apply only to newly designed airplanes. No manufacturer would commence production of a new model, unless sales were assured to be at least 2,000 for single-engine and 1,000 for multiengine models.

The benefits of the empennage fatigue resistance requirements will be the

prevention of fatalities, injuries and property damage caused by fatigue-related empennage structural parts failures. Estimating these benefits quantitatively is difficult because fatigue may have been a contributory factor in many accidents that were not directly attributed to fatigue failure.

Accident reports for a ten year period, 1972-82, were examined in order to determine the number of fatalities, injuries, and property damage attributable to empennage failures due to fatigue. Fourteen empennage fatigue-related accidents resulted in 21 fatalities, 2 serious and 4 minor injuries. In addition, 10 of the airplanes involved in these accidents were completely destroyed and the remaining 4 were severely damaged. Based on an assumed statistical value of \$1 million for each fatality, the FAA estimates that these accidents cost \$12,379,000 for single-engine aircraft and \$9,343,000 for twin-engine aircraft.

Estimating the number of accidents and their costs, which the rule would prevent in the future, is difficult because to do so requires predicting the number of aircraft subject to that rule that will join the fleet in the next 10 or 20 years. Therefore, in this analysis the FAA took an alternative approach to determine the cost effectiveness of the rule. This approach divides the above historic cost of empennage fatigue accidents by the number of aircraft in the fleet during the ten-year period in which they occurred, to obtain an average cost per aircraft for the period. This is assumed to be the average benefit per aircraft, if the rule were 100% effective in eliminating empennage fatigue accidents. This average potential benefit per aircraft was used to calculate the number of aircraft that would have to be manufactured in order for the "up front" cost of the rule to break even with the benefits that are expected to result from the rule. Table 2 shows that about 1,500 single-engine and 3,000 twin-engine aircraft would have to be manufactured to achieve this goal.

Since the start-up costs associated with a completely new model airplane are high and manufacturers must also cover product liability expenses as well as other non-manufacturing expenses, it is very likely that they will produce many more airplanes from a single model over a number of years than the 1,500 single-engine or 3,000 twin-engine airplanes necessary to have the benefits equal the cost of this rule.

Table 2.—Benefits and Costs Breakeven Point Analysis for Empennage Fatigue Regulations

	Type of Airplane	
	Single Engine	Twin Engine
Costs with load specs provided.....	91,788	986,901
Costs without load specs.....	102,662	1,000,882
Twenty year benefits.....	24,758,000	18,686,000
Annual benefits.....	1,237,900	934,300
Twenty year discount factors.....	8.514	8.514
Discounted benefits.....	10,539,481	7,954,630
Airplanes in present fleet.....	164,000	24,000
Discounted benefits per airplane.....	64.265	331.443
Number of airplanes required from any model so that benefits equal costs		
With load specs provided.....	1,428	2,978
Without load specs provided.....	1,597	3,020

Trade Impact Assessment

This rule will have little or no impact on trade opportunities for both U.S. firms doing business overseas and foreign firms doing business in the United States. Newly manufactured airplanes for the U.S. market, whether made by the United States or foreign manufacturers, will have to comply with the rule. Also, all newly certificated U.S. manufactured airplanes will have to meet the standards. The cost of compliance is minimal, ranging from perhaps a maximum of four percent of the cost of a new airplane to less than one percent of the cost. This added cost also creates a discernible benefit, making the airplanes which meet the standards a more attractive product, both in the United States and foreign markets.

Regulatory Flexibility Determination

The Regulatory Flexibility Act of 1980 (RFA) was enacted by Congress to ensure, among other things, that small entities are not disproportionately affected by government regulations. The RFA requires agencies to review rules which may have "a significant economic impact on a substantial number of small entities."

The FAA defines a small airplane manufacturer as one with fewer than 75 employees. It defines a substantial number as one that is not fewer than 11 and that is more than one-third of the small entities subject to the proposed rules. A significant economic impact is defined as \$14,258 per year in added

costs to each of these small manufacturers.¹

A review of the airplane manufacturers indicates that fewer than 11 "small" manufacturers will be subject to these regulations. Therefore, this rule does not have a significant economic impact on a substantial number of small airplane manufacturers.

Federalism Implications

The regulations adopted herein will not have substantial direct effects on the states, on the relationship between the national government and the states, or on the distribution of power and responsibilities among the various levels of government. Therefore, in accordance with Executive Order 12612, it is determined that this final rule does not have sufficient federalism implications to warrant the preparation of a Federalism Assessment.

Conclusion

The FAA has determined that this amendment is not a major rule under the provisions of Executive Order 12291, and is not significant under the provisions of the Department of Transportation Regulatory Policies and Procedures (44 FR 11034; February 26, 1979). For the reasons discussed earlier, the FAA certifies that the amendment will not have a significant economic impact, positive or negative, under provisions of the Regulatory Flexibility

Act. In addition, this amendment will have little or no impact on opportunity for U.S. firms doing business overseas or for foreign firms doing business in the United States. The evaluation prepared for this action is contained in the regulatory docket. A copy may be obtained by contacting the person listed under "FOR FURTHER INFORMATION CONTACT."

List of Subjects in 14 CFR Part 23

Aircraft, Air transportation, Aviation safety, Safety.

Adoption of the Amendment

Accordingly, part 23 of the Federal Aviation Regulations (14 CFR part 23) is amended as follows:

PART 23—AIRWORTHINESS STANDARDS: NORMAL, UTILITY, ACROBATIC, AND COMMUTER CATEGORY AIRPLANES

1. The authority citation for part 23 continues to read as follows:

Authority: 49 U.S.C. 1344, 1354(a), 1355, 1421, 1423, 1425, 1428, 1429, 1430; 49 U.S.C. 106(g) (Revised Pub. L. 97-449, January 12, 1983).

2. Section 23.572 is amended by revising the title, the introductory text of paragraph (a), and paragraphs (a)(1), and (b) to read as follows:

§ 23.572 Wing, empennage, and associated structures.

(a) The strength, detail design, and fabrication of those parts of the wings (including canards, tandem wings, and

winglets/tip fins), empennage, their carry-through and attaching structures, whose failure would be catastrophic, must be evaluated under either of the following unless it is shown that the structure, operating stress level, materials, and expected uses are comparable, from a fatigue standpoint, to a similar design that has had extensive satisfactory service experience:

(1) A fatigue strength investigation in which the structure is shown by analysis, tests, or both to be able to withstand the repeated loads of variable magnitude expected in service. Analysis alone is acceptable only when it is conservative and applied to simple structures; or

* * * * *

(b) Each evaluation required by this section must—

(1) Include typical loading spectra (e.g. taxi, ground-air-ground cycles, maneuver, gust);

(2) Account for any significant effects due to the mutual influence of aerodynamic surfaces; and

(3) Consider any significant effects from propeller slipstream loading, and buffet from vortex impingements.

Issued in Washington, DC, on September 21, 1989.

James B. Busey,
Administrator.

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¹ As defined in FAA Order 2100.14A, "Regulatory Flexibility Criteria and Guidance," effective September 16, 1986.